CMSE 801 - Day 15

Panda Data Anlysis Library

Oct 31, 2022

General Course Announcements

- Homework #3 is due Tuesday Nov 1 at 11:59 p.m.
- Homework #4 was released and is due Monday Nov 14 at 11:59 p.m.
- Homework #5 will be optional for extra credit
- We will talk about your project proposal next class

Grades

	Homework 1	Homework 2	Midterm
average	94%	95%	84%
median	96%	97%	90%
minimum	72%	78%	30%
maximum	100%	100%	100%

Questions/comments from Day 15 PCA

- As example is shown, when indexing first and second row of default dataframes by loc, the range is loc[0:1], but if index by iloc, the range is iloc[0:2, :], why is this different?
- I'm having a similar issue to when we first used matplotlib, that pandas is not a module
- As always, more practice will help to become more comfortable with dataframes
- Just going over indexing with pandas a lot more. I'm still getting tripped up by it a lot. But other than that it wasn't too bad.
- Can we add sub index in the panda dataframe as same as excel sheet?

In [1]:
```
import pandas as pd
set_example = pd.DataFrame([ ['A',1,2], ['B', 3, 4], ['C', 5, 6]])
print(set_example)
```

```
   0  1  2
0  A  1  2
1  B  3  4
2  C  5  6
```

In [2]:
```
set_example.index = set_example[0]
set_example = set_example.iloc[:,1:]
set_example.index.name = None
set_example.columns = ['c1', 'c2']
print(set_example)
```

```
   c1  c2
A   1   2
B   3   4
C   5   6
```

In [3]:
```python
print(set_example.iloc[0,0])
print("---")
print(set_example.iloc[1:3,0])
print("---")
print(set_example.iloc[1,0:2])
```

```
1
---
B    3
C    5
Name: c1, dtype: int64
---
c1    3
c2    4
Name: B, dtype: int64
```

In [4]:
```python
print(set_example['c1'].loc['A'])
print(set_example.loc['A','c1'])
print("---")
print(set_example.loc['B':'C','c1'])
print("---")
print(set_example.loc['B','c1':'c2'])
```

```
1
1
---
B    3
C    5
Name: c1, dtype: int64
---
c1    3
c2    4
Name: B, dtype: int64
```

In [5]:
```python
mask = set_example < 4
print(mask)
print(set_example[mask])
set_example[mask].describe()
```

```
      c1     c2
A   True   True
B   True  False
C  False  False
    c1   c2
A  1.0  2.0
B  3.0  NaN
C  NaN  NaN
```

Out[5]:

	c1	c2
count	2.000000	1.0
mean	2.000000	2.0
std	1.414214	NaN
min	1.000000	2.0
25%	1.500000	2.0
50%	2.000000	2.0

	c1	c2
75%	2.500000	2.0
max	3.000000	2.0